

6(a) (Student must draw at least 8 B-field lines.)



- (b) The coil experiences a magnetic force because it is placed in a magnetic field and the direction of current flow is not parallel to the magnetic field.
(can also say that the current is always perpendicular to the B-field)
- (c) Total length of the copper wire, $L = \text{No. of turns} \times \text{circumference}$
 $= N(2\pi r)$ (show equation)
 $= (250)(2\pi)(8.2 \times 10^{-3})$ (show substitution)
 $= \mathbf{12.88 \text{ m}}$ or $\mathbf{12.9 \text{ m}}$ (must give to 3 or 4 sf first)
 $\approx 13 \text{ m}$ [shown]
- (d) (Notice that the current is always perpendicular to the radial magnetic field.)
Magnetic force, $F = BIL \sin 90^\circ = BIL$
 $= (28 \times 10^{-3})(25 \times 10^{-3})(13)$
 $\therefore F = \mathbf{9.1 \times 10^{-3} \text{ N}}$
- or $F = BIL = (28 \times 10^{-3})(25 \times 10^{-3})(12.88) = 9.0 \times 10^{-3} \text{ N}$